ANY PROJECTIVE MODULE IS A DIRECT SUMMAND OF A FREE MODULE AFTER ADDING A FREE SUMMAND

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Eilenberg swindle: if F_0 = P plus Q is free then P plus a countable direct sum of copies of F_0 is isomorphic to that countable direct sum. invoked at proof lines 39005-39006. Provides the final isomorphism P plus F is isomorphic to F.

Source reference. Original lemma 0GVF: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38971–38971. Cited at proof lines 39005-3906 for the isomorphism P plus F is isomorphic to F; the isomorphism is load-bearing inside the counted span.

Lemma 0.2. *Let R be a ring. Let P be a projective module. There exists a free module F such that $P \oplus F$ is free.*

Proof. Since P is projective we see that $F_0 = P \oplus Q$ is a free module for some module Q. Set $F = \bigoplus_{n \geq 1} F_0$. Then $P \oplus F \cong F$ by Lemma 0GVF. □

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